Example 7.29

A thermometer was taken from a freezer and placed in a boiling water. It took 22 seconds for the thermometer to raise from $-10^\circ C$ to $100^\circ C$. Show that the rate of change of temperature at some time t is $5^\circ C$ per second.

Solution

Let $f(t)$ be the temperature at time t. By the mean value theorem, we have

$$f'(c) = \frac{f(b)-f(a)}{b-a}$$

$$= \frac{100-(-10)}{22}$$

$$= \frac{110}{22}$$

$$= 5^\circ C \text{ per second.}$$

Hence the instantaneous rate of change of temperature at some time t is $5^\circ C$ per second. ■

EXERCISE 7.3

1. Explain why Rolle's theorem is not applicable to the following functions in the respective intervals.

   (i) $f(x) = \left|\frac{1}{x}\right|,\ x \in [-1,1]$ (ii) $f(x) = \tan x, x \in [0,\pi]$

   (iii) $f(x) = x - 2\log x, x \in [2,7]$

2. Using the Rolle's theorem, determine the values of x at which the tangent is parallel to the x-axis for the following functions :

   (i) $f(x) = x^2 - x,\ x \in [0,1]$ (ii) $f(x) = \frac{x^2-2x}{x+2},\ x \in [-1,6]$

   (iii) $f(x) = \sqrt{x} - \frac{x}{3},\ x \in [0,9]$

3. Explain why Lagrange's mean value theorem is not applicable to the following functions in the respective intervals :

   (i) $f(x) = \frac{x+1}{x}, x \in [-1,2]$ (ii) $f(x) = |3x+1|,\ x \in [-1,3]$

4. Using the Lagrange's mean value theorem determine the values of x at which the tangent is parallel to the secant line at the end points of the given interval:

   (i) $f(x) = x^3 - 3x + 2,\ x \in [-2,2]$ (ii) $f(x) = (x-2)(x-7),\ x \in [3,11]$

5. Show that the value in the conclusion of the mean value theorem for

   (i) $f(x) = \frac{1}{x}$ on a closed interval of positive numbers $[a,b]$ is $\sqrt{ab}$

   (ii) $f(x) = Ax^2 + Bx + C$ on any interval $[a,b]$ is $\frac{a+b}{2}$.

6. A race car driver is in kilometer stone 20. If his speed never exceeds 150 km/hr, what is the maximum kilometer stone he can reach in the next two hours.

7. Suppose that for a function $f(x), f'(x) \le 1$ for all $1 \le x \le 4$. Show that $f(4) - f(1) \le 3$.

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